



Letter to the Editor

Response to “The Evidence Base for Electroconvulsive Therapy (ECT) in Geriatric Depression”

Dear Editor,

We are grateful to Dr. Kellner and colleagues for their thoughtful comments¹ on our study.² We fully agree with the points they raise and appreciate the opportunity to elaborate on our interpretation and provide additional information in line with their suggestions.

We concur that more patients would likely benefit from ECT if switching from unilateral to bilateral electrode placement after insufficient response were more routinely practiced. In our national cohort, 436 patients (83.5%) in the geriatric subgroup ($n = 522$) were initiated on brief-pulse ECT with unilateral electrode placement. However, only 13 patients (10.5%) of those who did not respond were switched to bilateral placement in an attempt to enhance efficacy. This suggests that many centers do not adhere to the Swedish clinical guideline for ECT, which recommends such a switch. It is worth noting, however, that this recommendation is primarily grounded in clinical

reasoning and experience rather than robust evidence from randomized trials of this specific situation.^{3,4}

Although the efficacy of ECT in geriatric depression is well established, there remains a striking lack of empirical answers to several critical clinical questions. For example, does the choice of brief pulse width impact efficacy or tolerability? Is determining electrical dose based on seizure threshold superior to using an age-adjusted high dose in unilateral ECT? When should a switch to bilateral electrode placement be considered? At what point is continued treatment unlikely to yield further benefit, and the series should be terminated?⁵ These are important questions that warrant further investigation through clinical trials and large-scale registry studies. We hope to contribute to this growing body of knowledge and encourage others to do the same.

In our view, the age-related differences observed in ECT response and comorbidity profiles among patients with major depressive disorder likely reflect underlying biological variation. These findings challenge the notion of major depressive disorder as a unified syndrome and raise questions about the validity of its current classification. More likely, major depressive disorder comprises of at least two

biologically distinct conditions, each with varying prevalence across the lifespan.⁶ We therefore suggest that future studies on the pathophysiology and treatment of MDD present their findings stratified by age group.

We also agree that the vast majority of geriatric patients tolerate the cognitive effects of ECT well. Swedish national registers provide opportunities similar to those available in Denmark, Scotland and a few other countries and we intend to explore whether the lack of association between ECT and subsequent dementia can be replicated in Swedish data.⁷

In conclusion, we share the view that the high response rates seen in our study and others support the central role of ECT in the treatment algorithm for severe geriatric depression.

Sincerely,
Tor Arnison and Axel Nordenskjöld

ChatGPT was used to improve the language in this manuscript.

DISCLOSURES

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Tor Arnison, Ph.D.¹

Axel Nordenskjöld, M.D., Ph.D.²

¹ *Clinical Epidemiology and Biostatistics, School of Medical Sciences, Faculty of Medicine and Health, Örebro University, Örebro, Sweden*

² *University Health Care Research Centre, Faculty of Medicine and Health, Örebro University, Örebro, Sweden*

References

1. Kellner CH, Sartorius A, Espinoza RT: The evidence base for electroconvulsive therapy (ECT) in geriatric depression. *Am J Geriatr Psychiatry* 2025; doi:[10.1016/j.jagp.2025.06.021](https://doi.org/10.1016/j.jagp.2025.06.021)
2. Arnison T, Eriksson A, Nordenskjöld A: Electroconvulsive therapy in the oldest-old patients with depression: response and remission rates, prognostic factors, adverse events and mortality. *Am J Geriatr Psychiatry* 2025; doi:[10.1016/j.jagp.2025.06.013](https://doi.org/10.1016/j.jagp.2025.06.013), S1064-7481(25)00381-1
3. Tew J.D. Jr, BH Mulsant, Haskett RF, et al: A randomized comparison of high-charge right unilateral electroconvulsive therapy and bilateral electroconvulsive therapy in older depressed patients who failed to respond to 5–8 moderate-charge right unilateral treatments. *J Clin Psychiatry* 2002; 63(12):1102–1105; doi:[10.4088/jcp.v63n1203](https://doi.org/10.4088/jcp.v63n1203)
4. Lapidus KA, Kellner CH: When to switch from unilateral to bilateral electroconvulsive therapy. *J ECT* 2011; 27(3):244–246; doi:[10.1097/YCT.0b013e31820059e1](https://doi.org/10.1097/YCT.0b013e31820059e1)
5. Cinar S, Oude Voshaar RC, Janzing JG, et al: The course of depressive symptoms in unipolar depressive disorder during electroconvulsive therapy: a latent class analysis. *J Affect Disord* 2010; 124(1–2):141–147; doi:[10.1016/j.jad.2009.11.002](https://doi.org/10.1016/j.jad.2009.11.002)
6. Shorter E: The doctrine of the two depressions in historical perspective. *Acta Psychiatr Scand Suppl* 2007; (433):5–13; doi:[10.1111/j.1600-0447.2007.00957.x](https://doi.org/10.1111/j.1600-0447.2007.00957.x)
7. Osler M, Rosing MP, Christensen GT, et al: Electroconvulsive therapy and risk of dementia in patients with affective disorders: a cohort study. *Lancet Psychiatry* 2018; 5(4):348–356; doi:[10.1016/S2215-0366\(18\)30056-7](https://doi.org/10.1016/S2215-0366(18)30056-7)